Semana 5: Representação Geométrica dos Complexos

O corpo $\mathbb{R}^2 = \{(a, b); a, b \in \mathbb{R}\}$, munido das operações

$$(a, b) + (c, d) = (a + c, b + d)$$

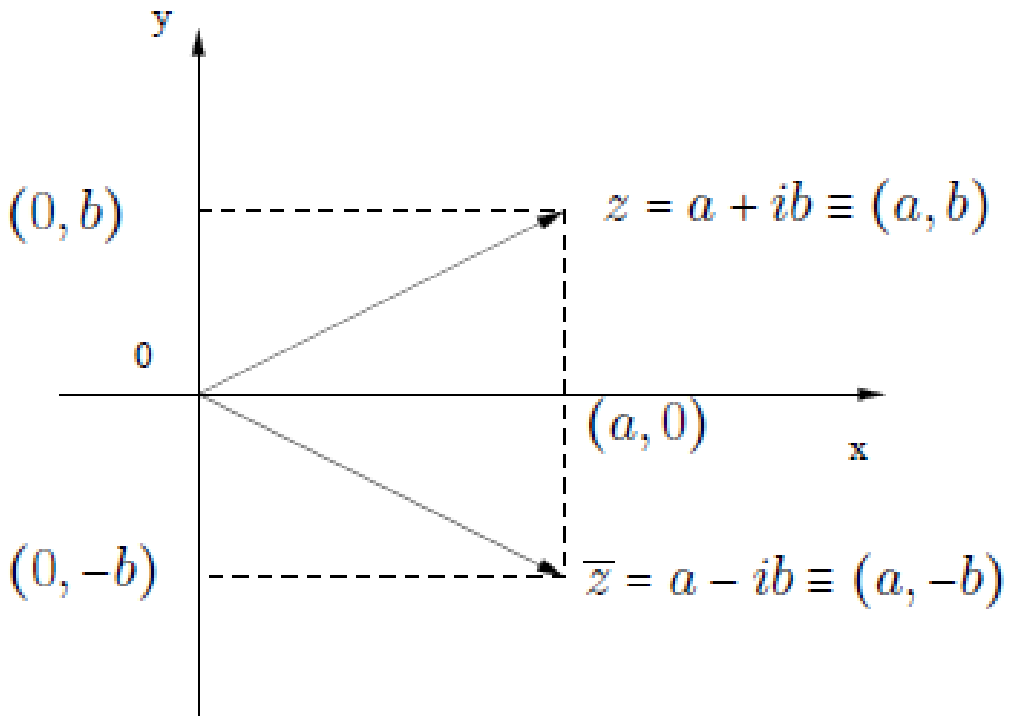
$$(a, b) \cdot (c, d) = (ac - bd, ad + bc)$$

é isomorfo ao conjunto dos números complexos $\mathbb{C}$, a partir do isomorfismo.

$$\varphi(a + bi) = (a, b)$$

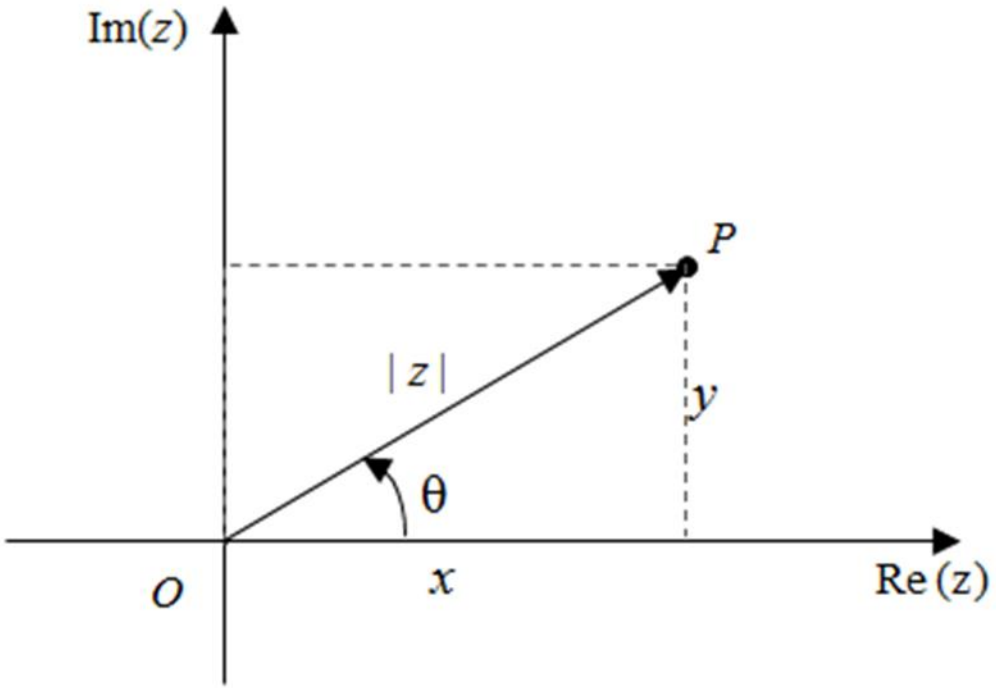
$$\lambda(a, b) = (\lambda, 0) \cdot (a, b) = (\lambda a - 0b, \lambda b + 0a) = (\lambda a, \lambda b)$$

$$\boldsymbol{z = a + bi = (a, b)}$$



$$|z| = \sqrt{a^2 + b^2}$$

Forma Polar



$$Re(z) = x = |z| \cos\theta$$

$$Im(z) = y = |z| \sin\theta$$

$$z = |z|(\cos\theta + i \sin\theta)$$

$$\bar{z} = |\bar{z}|(\cos(-\theta) + i \sin(-\theta)) = |z|(\cos\theta - i \sin\theta)$$

$\boldsymbol{z \in \mathbb{C}^*}$**:** $\boldsymbol{z = z(\rho, \theta) = \rho(\cos\theta + i \sin\theta)}$

Módulo de z: ρ

Argumento principal de z: θ

Argumentos de z: $\theta + 2k\pi, \ k \epsilon \mathbb{Z}$

FORMA ALGÉBRICA ⟶ FORMA POLAR

$z = -\sqrt{3} + i$

Passo 1: Determinar o módulo de z

$$\rho = |z| = \sqrt{\left(-\sqrt{3}\right)^2 + 1^2} = \sqrt{4} = 2$$

Passo 2: Determinar o argumento principal de z

$$\begin{cases} \cos\theta = \frac{Re(z)}{|z|} = \frac{-\sqrt{3}}{2} \\ \sin\theta = \frac{Im(z)}{|z|} = \frac{1}{2} \end{cases} \Longrightarrow \theta = \frac{5\pi}{6}$$

$$z = 2\left(\cos\frac{5\pi}{6} + i\sin\frac{5\pi}{6}\right)$$

PRODUTO E DIVISÃO NA FORMA POLAR

$$z_1 = \rho_1(\cos\theta_1 + i\sin\theta_1) \quad \text{e} \quad z_2 = \rho_2(\cos\theta_2 + i\sin\theta_2)$$

$$z_1 z_2 = \rho_1\rho_2(\cos\theta_1 + i\sin\theta_1)(\cos\theta_2 + i\sin\theta_2)$$

$$= \rho_1\,\rho_2[(\cos\theta_1\cos\theta_2 - \sin\theta_1\sin\theta_2) + i(\sin\theta_1\cos\theta_2 + \cos\theta_1\sin\theta_2)] = \rho_1\rho_2(\cos(\theta_1+\theta_2) + i\sin(\theta_1+\theta_2))$$

$$\frac{z_1}{z_2} = z_1 {z_2}^{-1} = z_1\frac{\overline{z_2}}{|z_2|^2} = \rho_1(\cos\theta_1 + i\sin\theta_1)\frac{\rho_2}{{\rho_2}^2}(\cos\theta_2 - i\sin\theta_2)$$

$$= \frac{\rho_1}{\rho_2}[(\cos\theta_1\cos\theta_2 + \sin\theta_1\sin\theta_2) + i(\sin\theta_1\cos\theta_2 - \cos\theta_1\sin\theta_2)] = \frac{\rho_1}{\rho_2}(\cos(\theta_1-\theta_2) + i\sin(\theta_1-\theta_2))$$

Para multiplicar dois complexos na forma polar, multiplicamos os módulos e somamos os argumentos.

Para dividir dois complexos na forma polar, dividimos os módulos e subtraímos os argumentos

FÓRMULA DE DE MOIVRE

Potência:

$$\begin{cases} z^0 = 1 \\ z^n = z \cdot z^{n-1}, n \in \{1,2,3,\dots\} \\ z^n = (z^{-1})^{-n}, n \in \{-1,-2,-3,\dots\} \end{cases}$$

$$z = \rho(\cos\theta + i\sin\theta) \Longrightarrow z^n = \rho^n(\cos(n\theta) + i\sin(n\theta)), n \in \mathbb{N}$$

INDUÇÃO:

$n = 0$: $\quad z^0 = \rho^0(\cos(0\theta) + i\sin(0\theta)) = 1 + 0i = 1$

$n \Longrightarrow n + 1$:

$$\begin{aligned} z^{n+1} = z \cdot z^n &= \rho(\cos\theta + i\sin\theta)\rho^n(\cos(n\theta) + i\sin(n\theta)) \\ &= \rho^{n+1}[(\cos\theta\cos(n\theta) - \sin\theta\sin(n\theta)) + i(\cos\theta\sin(n\theta) + \sin\theta\cos(n\theta))] \\ &= \rho^{n+1}(\cos(\theta + n\theta) + i\sin(\theta + n\theta)) = \rho^{n+1}(\cos((n+1)\theta) + i\sin((n+1)\theta)) \end{aligned}$$

FÓRMULA DE DE MOIVRE

$$z = \rho(\cos\theta + i\sin\theta) \Longrightarrow z^n = \rho^n(\cos(n\theta) + i\sin(n\theta)), n \in \{-1, -2, -3, \dots\}$$

$n = -m,$ com $m \in \mathbb{N}$

$$z^n = (z^{-1})^{-n} = (z^{-1})^m = \left(\frac{\bar{z}}{|z|^2}\right)^m = \left(\frac{\rho}{\rho^2}(\cos\theta - i\sin\theta)\right)^m = \left(\frac{1}{\rho}(\cos(-\theta) + i\sin(-\theta))\right)^m$$

$$= \frac{1}{\rho^m}(\cos(-m\theta) + i\sin(-m\theta)) = \rho^{-m}(\cos(-m\theta) + i\sin(-m\theta)) = \rho^n(\cos(n\theta) + i\sin(n\theta))$$